Huawei Access Node Fault Resolution SOP (BAS-ACC-07 - Basavanagudi POP)

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_5575

Description

This Standard Operating Procedure outlines the diagnostic and resolution steps for a hardware/environmental fault occurring on the Huawei Access Node (BAS-ACC-07) at the Basavanagudi POP, manifested by ASIC memory parity errors, line card crashes, or power supply redundancy loss, as indicated by the ALARM_5575 code. The severity of this fault is classified as Minor. The Huawei Access Node is a critical component in the telecom network infrastructure, responsible for providing broadband access services to subscribers. The fault is likely caused by a hardware issue, such as a failed ASIC chip or a power supply failure, which can be attributed to environmental factors like high temperature, humidity, or power quality issues. The failure domain is isolated to the Access role, where the Huawei Access Node operates. The diagnostic and resolution steps outlined in this SOP aim to identify and rectify the root cause of the fault, ensuring minimal downtime and maintaining the overall network availability.

Pre-requisite Checks

- Verify the Basavanagudi POP site location and confirm the presence of the Huawei Access Node (BAS-ACC-07).

- Ensure remote hands are dispatched to the site to provide physical access and assistance as needed.

- Check the redundant link status and confirm that the affected link is indeed redundant and not the primary link.

- Verify the power supply redundancy status and confirm that the power supply is functioning properly.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display fault-management fault, Expected Output: Fault ID: ALARM_5575, Severity: Minor, Description: This command displays the fault management information, which should indicate the ALARM_5575 code and Minor severity. This command helps identify the specific fault and its severity.
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Command: display hardware all, Expected Output: ASIC memory parity error detected, Description: This command displays the hardware information, which should indicate the ASIC memory parity error. This command helps identify the potential root cause of the fault.

Command: display power-supply all, Expected Output: Power supply redundancy loss detected, Description: This command displays the power supply information, which should indicate the power supply redundancy loss. This command helps identify the potential root cause of the fault.
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Resolution Steps

Step 1: Step: Replace the faulty ASIC chip, Command: system replace asic slot 1, Description: This command initiates the replacement of the faulty ASIC chip in slot 1. This step should rectify the ASIC memory parity error and restore the node to normal operation.

Step 2: Step: Replace the faulty line card, Command: system replace line-card slot 2, Description: This command initiates the replacement of the faulty line card in slot 2. This step should rectify the line card crash and restore the node to normal operation.

Step 3: Step: Check and replace the power supply unit (PSU), Command: system replace power-supply slot 3, Description: This command initiates the replacement of the power supply unit (PSU) in slot 3. This step should rectify the power supply redundancy loss and restore the node to normal operation.

Post-Check Verification

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Command: display fault-management fault, Expected Output: Fault ID: None, Severity: None, Description: This command verifies that the fault has been cleared and the node is operating normally. If the fault still persists, further diagnostic steps may be required.

Command: display hardware all, Expected Output: No hardware issues detected, Description: This command verifies that the hardware is functioning properly and no issues are detected. If any issues are still present, further diagnostic steps may be required.

Command: display power-supply all, Expected Output: Power supply redundancy is functioning properly, Description: This command verifies that the power supply redundancy is functioning properly and no issues are detected. If any issues are still present, further diagnostic steps may be required.
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Roll-back Procedures

- Step: Restore the original configuration, Command: system restore original, Description: This command restores the original configuration of the node, which should revert any changes made during the resolution process. This step ensures that the node returns to its original state and function.

- Step: Reboot the node, Command: system reboot, Description: This command reboots the node, which should restore it to a stable state. This step ensures that the node is functioning properly and any issues are resolved.